

Endpoint sharpness in two Grand Lebesgue inequalities via concentration and Gaussian dilations

Candidate partial result from run `fa_banach_001`, lane 18

11 August 2026

Abstract

Lifyand, Ostrovsky, and Sirota left open the sharpness of a Sobolev trace estimate and a Young convolution estimate in bilateral Grand Lebesgue spaces. Their fixed borderline examples lose respectively $1/n$ and one unit in the endpoint blow-up exponent. We show that these losses are artifacts of using one fixed input. Normalized smooth concentration families and normalized Gaussian families attain, up to constants, the complete endpoint orders predicted by the two source inequalities for two-sided power gauges. The same argument permits positive-index slowly varying endpoint modifiers. Exact constants and completely arbitrary generating functions are not addressed.

1 Source question and scope

For a positive generating function ψ on (a, b) , write

$$\|f\|_{G(\psi; a, b)} := \sup_{a < p < b} \frac{\|f\|_{L^p}}{\psi(p)}.$$

The source paper arXiv:0801.3404 proves a Sobolev trace estimate, labelled (in2), and a convolution estimate, labelled (in3). In its final Sharpness section it says (source PDF p. 11):

Finding sharp estimates in these two cases is an interesting open problem.

For the power gauge

$$\psi_{a,b}^{\alpha, \beta}(p) := (p - a)^{-\alpha} (b - p)^{-\beta}, \quad a < p < b, \quad (1)$$

the source's fixed examples give the correct endpoint interval but fall short of the upper estimate by $1/n$ in (in2) and by 1 in (in3).

The result below closes both exponent gaps in the natural operator-norm sense: the test function may depend on the exponent tending to the endpoint. We work with the standard restriction trace on $\mathbb{R}^m \subset \mathbb{R}^n$ and with convolution on $\mathbb{R}^d$. These geometries already prove that the source estimates cannot be improved in general.

2 Proof intuition

A fixed borderline singularity mixes growth of its input norm with the smoothing caused by restriction or convolution. It therefore need not measure the operator norm near every endpoint. Instead, move a smooth profile through its dilation orbit.

For the trace inequality, $u_t(x) = c_t \phi(tx)$ has

$$\|\nabla u_t\|_p = c_t t^{1-n/p} \|\nabla \phi\|_p, \quad \|u_t\|_{\mathbb{R}^m} = c_t t^{-m/q} \|\phi\|_{\mathbb{R}^m}.$$

Normalizing the first expression in the Grand Lebesgue norm introduces $(\log t)^\beta$ as $t \rightarrow \infty$. The remaining power of t is $\exp[-m(1/q - 1/B) \log t]$. Choosing $\log t \asymp (B - q)^{-1}$ recovers the missing endpoint power.

For convolution, Gaussians are closed under dilation and convolution. Their normalizers contribute $(\log t)^{\beta_1 + \beta_2}$, while their convolved L^r norm contributes $\exp[-(d/2)(1/r - 1/B) \log t]$. The same optimization recovers the whole predicted exponent. Broad Gaussians give the lower endpoints.

3 A normalizer lemma

We use $F \asymp G$ for two-sided estimates with constants independent of the parameter tending to the indicated endpoint.

Lemma 1 (endpoint Laplace principle). *Let $0 < a < b < \infty$, $\alpha, \beta > 0$, and let c be positive and continuous on $[a, b]$. Put*

$$M_\rho(t) = \sup_{a < p < b} c(p) t^{\rho - \kappa/p} (p - a)^\alpha (b - p)^\beta, \quad \kappa > 0.$$

Then

$$\begin{aligned} M_\rho(t) &\asymp t^{\rho - \kappa/b} (\log t)^{-\beta}, & t \rightarrow \infty, \\ M_\rho(t) &\asymp t^{\rho - \kappa/a} (\log(1/t))^{-\alpha}, & t \downarrow 0. \end{aligned}$$

Proof. For $t = e^L \rightarrow \infty$, set $s = b - p$. After factoring out $t^{\rho - \kappa/b}$, the remaining exponential is

$$\exp \left[-\kappa L \left(\frac{1}{p} - \frac{1}{b} \right) \right] = \exp \left[-\frac{\kappa L s}{bp} \right].$$

On a fixed neighborhood of b , all other factors except s^β are bounded above and below. Thus the supremum is comparable to $\sup_{0 < s < s_0} s^\beta e^{-\kappa L s} \asymp L^{-\beta}$. Outside that neighborhood it is exponentially smaller. The proof at zero is identical, with $s = p - a$ and $L = \log(1/t)$. $\square$

4 Sharp Sobolev trace endpoints

Let $Tu = u|_{\mathbb{R}^m}$ be restriction to $\mathbb{R}^m \times \{0\} \subset \mathbb{R}^n$. Fix

$$1 \leq a < b < n, \quad A = \frac{ma}{n-a}, \quad B = \frac{mb}{n-b}, \quad (2)$$

and assume $1 \leq A < B$. Let $p(q) = qn/(q+m)$, so $p(A) = a$, $p(B) = b$, and define the source target gauge

$$\nu(q) = q^{1-1/n} \psi_{a,b}^{\alpha,\beta}(p(q)). \quad (3)$$

Theorem 1 (both trace endpoint exponents are sharp). *For*

$$\mathcal{S}(q) := \sup \left\{ \|Tu\|_{L^q(\mathbb{R}^m)} : u \in C_c^\infty(\mathbb{R}^n), \|\nabla u\|_{G(\psi_{a,b}^{\alpha,\beta})} \leq 1 \right\},$$

the source upper estimate and matching lower estimates give

$$\begin{aligned} \mathcal{S}(q) &\asymp \nu(q) \asymp (q - A)^{-\alpha}, & q \downarrow A, \\ \mathcal{S}(q) &\asymp \nu(q) \asymp (B - q)^{-\beta}, & q \uparrow B. \end{aligned}$$

Consequently neither endpoint power in (in2) can be decreased.

Proof. The upper bounds are (in2), and $p'(q) = nm/(q + m)^2 > 0$ shows that $p(q) - a \asymp q - A$ and $b - p(q) \asymp B - q$.

Choose $\phi \in C_c^\infty(\mathbb{R}^n)$ with $T\phi \neq 0$. Put $u_t = c_t \phi(t \cdot)$ and choose c_t so that $\|\nabla u_t\|_{G(\psi)} = 1$. Since

$$\|\nabla(\phi(t \cdot))\|_p = t^{1-n/p} \|\nabla \phi\|_p,$$

Lemma 1, with $\rho = 1$ and $\kappa = n$, gives

$$\begin{aligned} c_t &\asymp t^{n/b-1} (\log t)^\beta, & t \rightarrow \infty, \\ c_t &\asymp t^{n/a-1} (\log(1/t))^\alpha, & t \downarrow 0. \end{aligned}$$

Also

$$\|Tu_t\|_q = c_t t^{-m/q} \|T\phi\|_q,$$

where $\|T\phi\|_q$ stays bounded above and below on $[A, B]$. The identities $n/b - 1 = m/B$ and $n/a - 1 = m/A$ yield

$$\begin{aligned} \|Tu_t\|_q &\asymp t^{-m(1/q-1/B)} (\log t)^\beta, & t \rightarrow \infty, \\ \|Tu_t\|_q &\asymp t^{m(1/A-1/q)} (\log(1/t))^\alpha, & t \downarrow 0. \end{aligned}$$

For $q < B$, choose $\log t = (1/q - 1/B)^{-1}$. For $q > A$, choose $\log(1/t) = (1/A - 1/q)^{-1}$. The reciprocal differences are comparable to $(B - q)^{-1}$ and $(q - A)^{-1}$, proving the lower bounds. $\square$

5 Sharp convolution endpoints

For $i = 1, 2$, let

$$\psi_i(p) = (p - a_i)^{-\alpha_i} (b_i - p)^{-\beta_i}, \quad 1 \leq a_i < b_i < \infty, \quad \alpha_i, \beta_i > 0.$$

Assume

$$\frac{1}{a_1} + \frac{1}{a_2} > 1, \quad \frac{1}{b_1} + \frac{1}{b_2} > 1,$$

and define

$$\frac{1}{A} = \frac{1}{a_1} + \frac{1}{a_2} - 1, \quad \frac{1}{B} = \frac{1}{b_1} + \frac{1}{b_2} - 1. \quad (4)$$

For $A < r < B$, let

$$\tau(r) = \inf_{\substack{a_1 < p < b_1, \ a_2 < q < b_2 \\ 1/p + 1/q = 1 + 1/r}} \psi_1(p) \psi_2(q), \quad (5)$$

the target gauge in (in3).

Theorem 2 (both convolution endpoint exponents are sharp). *Let*

$$\mathcal{C}(r) = \sup \left\{ \|f * g\|_{L^r(\mathbb{R}^d)} : \|f\|_{G(\psi_1)} \leq 1, \|g\|_{G(\psi_2)} \leq 1 \right\}.$$

Then

$$\begin{aligned} \mathcal{C}(r) &\asymp \tau(r) \asymp (r - A)^{-(\alpha_1 + \alpha_2)}, & r \downarrow A, \\ \mathcal{C}(r) &\asymp \tau(r) \asymp (B - r)^{-(\beta_1 + \beta_2)}, & r \uparrow B. \end{aligned}$$

Thus the source's one-unit fixed-function gap does not occur for the bilinear operator norm.

Proof. The upper bound $\mathcal{C}(r) \leq \tau(r)$ is (in3). Near B , write $p = b_1 - s_1$ and $q = b_2 - s_2$. The constraint (5) is exactly

$$\frac{s_1}{b_1 p} + \frac{s_2}{b_2 q} = \frac{1}{r} - \frac{1}{B}. \quad (6)$$

Thus $s_1 + s_2 \asymp B - r$. Minimizing $s_1^{-\beta_1} s_2^{-\beta_2}$ subject to (6) gives $\tau(r) \asymp (B - r)^{-(\beta_1 + \beta_2)}$. Near A , use $p = a_1 + s_1$, $q = a_2 + s_2$, and $1/A - 1/r \asymp r - A$.

For the lower bounds put $h_t(x) = e^{-t|x|^2}$ and $N_i(t) = \|h_t\|_{G(\psi_i)}$. Since

$$\|h_t\|_p = \left(\frac{\pi}{pt} \right)^{d/(2p)},$$

Lemma 1 gives

$$\begin{aligned} N_i(t) &\asymp t^{-d/(2b_i)} (\log t)^{-\beta_i}, & t \rightarrow \infty, \\ N_i(t) &\asymp t^{-d/(2a_i)} (\log(1/t))^{-\alpha_i}, & t \downarrow 0. \end{aligned}$$

The functions $f_t = h_t/N_1(t)$ and $g_t = h_t/N_2(t)$ have norm one. The exact identity

$$h_t * h_t(x) = \left(\frac{\pi}{2t} \right)^{d/2} e^{-t|x|^2/2}$$

implies

$$\begin{aligned} \|f_t * g_t\|_r &\asymp t^{-\frac{d}{2}(1/r - 1/B)} (\log t)^{\beta_1 + \beta_2}, & t \rightarrow \infty, \\ \|f_t * g_t\|_r &\asymp t^{\frac{d}{2}(1/A - 1/r)} (\log(1/t))^{\alpha_1 + \alpha_2}, & t \downarrow 0. \end{aligned}$$

Choose $\log t = (1/r - 1/B)^{-1}$ in the first line and $\log(1/t) = (1/A - 1/r)^{-1}$ in the second. This gives the asserted lower orders. $\square$

6 Regularly varying upgrade

The mechanism is not restricted to pure powers. Suppose near an upper endpoint that

$$\psi(p) \asymp (b - p)^{-\beta} \ell\left(\frac{1}{b - p}\right), \quad \beta > 0, \quad (7)$$

where ℓ is positive, locally bounded together with its reciprocal, and slowly varying at infinity. Potter bounds in Lemma 1 give an extra factor $1/\ell(\log t)$ in the normalizer. Therefore

$$\mathcal{S}(q) \asymp (B - q)^{-\beta} \ell\left(\frac{1}{B - q}\right).$$

For convolution, the two slowly varying factors multiply:

$$\mathcal{C}(r) \asymp \tau(r) \asymp (B-r)^{-(\beta_1+\beta_2)} \ell_{1,+} \left(\frac{1}{B-r} \right) \ell_{2,+} \left(\frac{1}{B-r} \right).$$

The lower endpoints are analogous. Positive power index keeps the maximizing dilation at distance comparable to $1/\log t$ from the endpoint; index-zero gauges can require a different conjugate scale.

7 Verification, novelty, and limitations

The proof uses exact dilation identities, the elementary Gaussian convolution formula, and the source upper estimates. Every exponent follows from

$$\frac{n}{b} - 1 = \frac{m}{B}, \quad \frac{n}{a} - 1 = \frac{m}{A}, \quad \frac{1}{B} = \frac{1}{b_1} + \frac{1}{b_2} - 1, \quad \frac{1}{A} = \frac{1}{a_1} + \frac{1}{a_2} - 1.$$

The optimizing scales tend in the required directions. No numerical or symbolic computation is used.

A search by the exact source title and open-problem phrase, and by Grand Lebesgue Sobolev/convolution sharpness keywords, did not locate these endpoint lower bounds. The later interpolation paper arXiv:1705.06170 gives Young-type upper estimates in related spaces but does not supply this lower-bound argument. This should be regarded as a candidate new partial theorem pending expert literature review.

The packet does not determine exact constants, treats the trace in the Euclidean model geometry, and treats convolution on $\mathbb{R}^d$ rather than every unimodular group. Arbitrary generating functions without regular endpoint behavior can be governed by a different Legendre envelope. Human review is recommended, especially for the source's intended projection convention and for possible overlap with limiting interpolation results.

References

- [1] E. Liflyand, E. Ostrovsky, and L. Sirota, *Tensor, Sobolev, Multiplicative and Convolution Operators in the Bide-Side Grand Lebesgue Spaces*, arXiv:0801.3404 (2008).
- [2] H. J. Brascamp and E. H. Lieb, *Best constants in Young's inequality, its converse, and its generalization to more than three functions*, J. Funct. Anal. 20 (1976), 151–173.
- [3] P. Fernández-Martínez and E. Brandani da Silva, *New Young inequalities and applications*, arXiv:1705.06170; Z. Anal. Anwend. 38 (2019), 419–437.